THE EVENT-STATE PROTOCOLS

A Unified Explanatory Framework for Reality

Torben Wille
System Architect & Independent Researcher

November 2025

Abstract

This monograph explores the systemic implications of **Event-State Theory (EST)**. By defining the universe as a discrete computational optimization process governed by a thermodynamic Cost Function, this framework offers novel resolutions to foundational paradoxes. It unifies the macro-scale geometry of Cosmology with the micro-scale mechanics of Quantum Dynamics, proposing that "Time" is an emergent property of asynchronous state selection. This document maps the applicability of EST across Fundamental Physics, Material Science, and Complex Systems.

Contents

Chapter 1

The Operating System: Axioms of EST

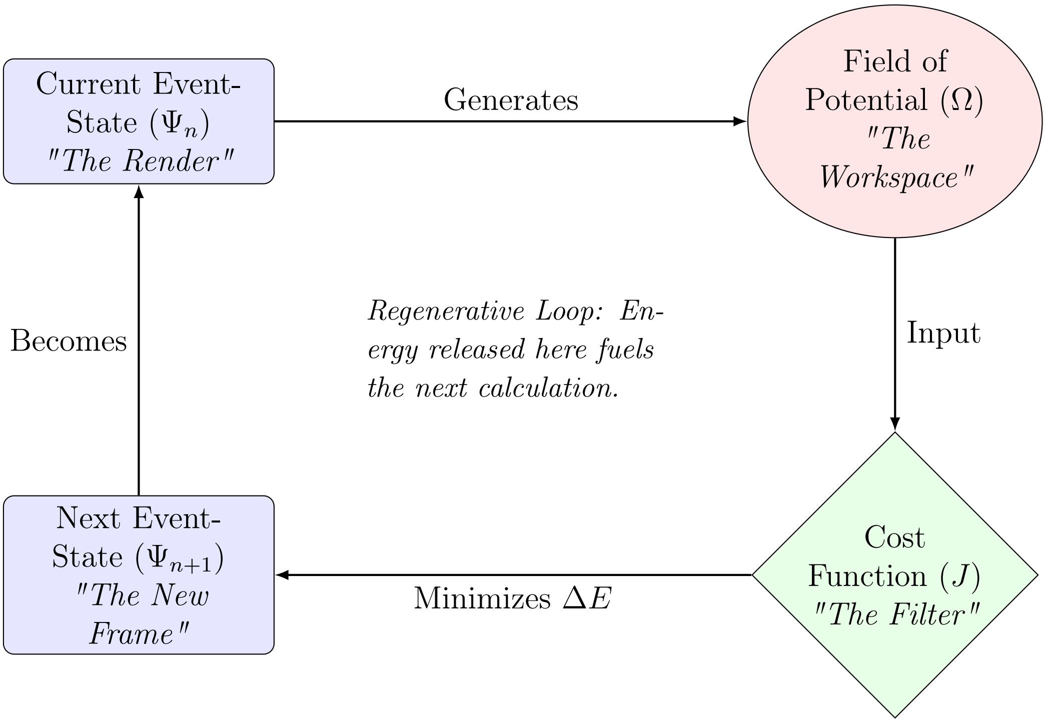

Figure 1.1: **The Event-State Computational Cycle.** The universe iterates by generating a field of potential from the current state, filtering it through the Cost Function (J), and crystallizing the path of least resistance into the next frame.

Redefining the Fundamental Constants.

The Failure of Continuum Physics

Why the assumption of continuous spacetime creates singularities and paradoxes (e.g., Zeno's Paradox, Renormalization).

Axiom I: Time as Frame Rate (n)

The Emergence of Relativity from Asynchronicity

Reconciling the Global State with Local Causality.

A common critique of discrete frame theories is the apparent violation of Lorentz Invariance (the lack of a preferred reference frame). EST resolves this by defining the "Global State" (Ψ) not as a flat geometric hypersurface, but as a **Ragged Computational Wavefront**.

1.3.1 The "Ragged Now"

Under Asynchronous Multi-Point Crystallization (AMPC), the universe does not update simultaneously. Different regions possess different local Frame Counts (n). The "Present" is a jagged processing front, similar to a fire line moving through a forest. There is no single "Now"; there is only a local update sequence.

1.3.2 Relativity as Network Latency

Since the Cost Function (J) is calculated locally based on immediate neighbors in the causal graph, information propagation is limited by the connectivity of the nodes.

- **The Speed of Light (c):** Defined as the maximum propagation velocity of the "Update Signal" through the discrete lattice.
- **Lorentz Invariance:** Emerges as the effective description of this propagation limit for any observer *inside* the system. An observer cannot measure the "Server Tick Rate" (Universal Frame Rate) because their own consciousness is running on that same clock.

Conclusion: Relativity is not a fundamental property of space; it is the "User Interface" of an asynchronous distributed network.

Axiom II: The Field of Potential (Ω)

Moving beyond the "Empty Vacuum." Defining space as a high-density field of potential information. Physical matter is the "rendered" output; the vacuum is the processing substrate.

Axiom III: The Cost Function (J)

Formalizing the Cost: Kolmogorov Complexity

From Metaphor to Mathematics.

To transition EST from a conceptual framework to a physical theory, we must quantify the concept of "Simplicity." We propose that the Cost Function (J) is homologous to **Conditional Kolmogorov Complexity** (K).

1.6.1 The Principle of Computational Least Action

(Formerly: The Algorithmic Probability Rule)

Standard physics is governed by the Principle of Least Action (minimizing the Lagrangian). EST generalizes this to the **Principle of Computational Least Action**. The probability of the universe selecting a specific successor state ψ is inversely proportional to the length of the shortest algorithm required to derive ψ from the current state Ψ_n.

$$P(\Psi_{n+1} = \psi) \propto \exp\left(-\frac{K(\psi|\Psi_n)}{\lambda}\right)$$

Where:

- $K(\psi|\Psi_n)$ is the computational cost (in bits) to render the new frame given the old frame.
- λ is the "Causal Temperature" (or processing bandwidth) of the system.

1.6.2 Application to Cosmology

This formalism provides the mechanism for the Hubble Tension solution (AMPC).

- **Voids (Low Complexity):** A region of empty space is algorithmically compressible. The update cost K is low. The effective Frame Rate (τ) is high. Result: Faster Expansion.
- **Galaxies (High Complexity):** A region of dense matter is algorithmically incompressible. The update cost K is high. The effective Frame Rate (τ) is throttled. Result: Slower Expansion.

Replacing Randomness with Economy.

$$J(\Psi_n \to \psi) = \alpha \Delta E + \beta(1 - I_F) + \epsilon$$

The universe selects the next frame by minimizing Thermodynamic Cost (ΔE) and maximizing Information Fidelity (I_F).

Clarifying the Substrate: Topological Constants

Defining the Limits of Propagation.

To reconcile the discrete nature of EST with the observed success of relativity, we must define the fundamental constraints of the causal graph.

1.7.1 The Connectivity Constant ($c \equiv 1$)

The "Speed of Light" is not a fundamental velocity in meters per second; it is a topological constraint of the network. Information propagates at exactly **one causal adjacency per frame update** ($c \equiv 1$).

- **Implication:** Light cannot "teleport" across the graph; it must traverse neighbor-to-neighbor.
- **Result:** Lorentz Invariance is the emergent description of this propagation limit for any observer *inside* the simulation.

1.7.2 The Locality of Optimization

The Cost Function (J) is not a centralized "God's Eye" calculation. It is strictly local.

- **Decentralized Processing:** Each node selects its future state based *solely* on the Field of Potential (Ω) of its immediate causal neighbors.
- **Emergence:** The apparent "Global Optimization" of the universe is the aggregate result of trillions of local optimizations occurring in a chain reaction. There is no central processor; there is only the collective mesh.

Chapter 2

Cosmology: The Hardware Architecture

Solving the Crisis of the Macro-Scale.

The Hubble Tension Resolution (AMPC)

The Anomaly: The discrepancy between Early ($z > 1000$) and Late ($z < 1$) measurements of the Hubble Parameter (H_0).

The EST Solution: Asynchronous Multi-Point Crystallization. The Big Bang was not a singular event but a staggered nucleation process. The observed tension is a measurement of the geometric interference between our local nucleation bubble and surrounding domains.

2.1.1 Dimensional Definition of the Expansion Field

In the formal proposition, the local expansion rate $H(x)$ is modeled as a superposition of radial fields from nucleation centers C_i:

$$H(x) \approx \sum_i \frac{A_i}{|x - C_i|}$$

To satisfy dimensional analysis where $[H] = T^{-1}$ (Frequency) and $[x] = L$ (Distance), the amplitude term A_i is defined as the **Causal Propagation Rate** ($[A] = L \cdot T^{-1}$).

- Physically, A_i represents the **Growth Velocity** of the nucleation bubble (Nodes per Frame).
- A high-A region indicates a domain where the "Update Wave" propagates rapidly through the causal graph (Vacuum/Void).
- A low-A region indicates a domain with high computational drag (Filament/Matter).

Note on Formalism: This dimensional definition resolves the ambiguity present in the initial manuscript (Zenodo Record 17698703) and establishes the rigorous basis for the forthcoming Version 2 of the formal theory.

Dark Matter as Gravitational Potential

Galactic rotation curves indicate that visible matter accounts for only 15% of the required gravity to hold galaxies together. The standard model inserts a hidden particle (Dark Matter) to fill the gap.

In EST, Gravity is defined not as a force generated by mass, but as the "Inertial Grip" of the Field of Potential (Ω) calculating the next frame. A complex system like a galaxy possesses a massive cloud of potential future states. This high-density information field exerts a "Causal Pressure" on the present frame, pulling stars inward. We are not detecting missing mass; we are detecting the weight of the galaxy's future.

Dark Energy as Vacuum Pressure

The accelerating expansion of the universe is typically attributed to a Cosmological Constant (Λ). EST reinterprets this as a boundary condition. As the crystallized universe ($I_F \approx 1$) expands into the infinite Field of Potential ($I_F \approx 0$), the Cost Function minimizes resistance by accelerating the "rendering rate" at the periphery. Dark Energy is the vacuum pressure of the system processing the infinite into the finite.

Black Holes: Information Saturation & The Stalled Frame

Redefining the Event Horizon as a Computational Limit.

Standard General Relativity predicts a singularity of infinite density. EST predicts a singularity of infinite *processing latency*.

2.4.1 The Stalled Render

A Black Hole is a region where local Information Density (ρ_{info}) reaches a critical threshold. The computational cost (ΔE) required to calculate the next state becomes astronomically high. Consequently, the local Frame Rate (τ_{local}) approaches zero. The system effectively "hangs." Matter is not destroyed; it is locked in a static, un-rendered state.

2.4.2 Hawking Radiation as Informational Catapulting

The Information Paradox is resolved by viewing Hawking Radiation as a "System Reboot" via vacuum fluctuations.

- **The Mechanism:** A quantum fluctuation in the Field of Potential (Ω) injects a micro-packet of energy (ΔE) at the Event Horizon.
- **The Catapult:** This external energy pays the "Cost" that the Black Hole could not pay itself. It forces a single bit of locked information to transition from the "Stalled Frame" back into the active Field of Potential.
- **Result:** Information is not lost; it is sequentially "catapulted" back into the universe, bit by bit.

Chapter 3

Quantum Dynamics: The Micro-Kernel

Reframing Uncertainty and Causality.

The Geometric Collapse of the Wave Function

Deriving the Born Rule from Cost Optimization.

Standard Quantum Mechanics posits that probability is determined by the square of the wavefunction amplitude ($P = |\psi|^2$), but offers no mechanism for *why* this relation holds. EST proposes a thermodynamic origin.

3.1.1 The Rosetta Stone Equation

The wavefunction $\Psi(x,t)$ is reinterpreted as a description of the **Field of Potential (Ω)**. The probability of transition to a state ψ is governed by the EST Selection Rule:

$$\exp\left(-\frac{J(\Psi_n \to \psi)}{\lambda}\right) \propto |\langle\psi|\hat{U}|\Psi_n\rangle|^2$$

Here, the quantum interference pattern is not the result of wave mechanics, but of the Cost Function (J) seeking the global minimum of Information Fidelity (I_F). Constructive interference represents a "Low Cost" path; destructive interference represents a "High Cost" path.

3.1.2 Subsuming General Relativity

Similarly, the curvature of spacetime described by Einstein's Field Equations is identified as the **Equilibrium Condition** for macroscopic information transfer. Gravity is the geometric configuration that minimizes the energy cost (ΔE) of updating high-density information states.

The "Reverse Splash" Mechanism

The Hypothesis: A collision event ($t = 0$) is triggered by the convergence of potential waves ($t < 0$).

- **Pre-Causal Resonance:** Detecting the "inward rush" of potential before the event occurs.
- **The Geometric Collapse:** The particle is located where the ripples intersect.

Causal Friction & High-Energy Anomalies

The Anomaly: "Jet Quenching" and the "Ridge Effect" at the LHC.
The EST Solution: This 'drag' occurs because the outgoing real signal (the jet) and the post-causal component of the pre-causal resonance are **out of phase**. This leads to **destructive interference** at the wavefront, creating an 'Informational Headwind' that robs the jet of initial momentum, mimicking the viscosity of a dense medium (QGP) without requiring a fluid.

Baryon Asymmetry as "Scar Tissue"

This mechanism is implemented mathematically by the **Primordial Asymmetry Constant** (ϵ) within the Cost Function. The term ϵ_{MM} imposes a consistent, microscopic bias where $J(\psi_{Matter}) < J(\psi_{Antimatter})$. Consequently, matter-dominant outcomes are systematically 'cheaper' to render across all frames, ensuring the persistence of structure over annihilation.

Chapter 4

Complex Systems & The Living Frame

Implications for Biology and Sociology.

The Fermi Paradox: The Optimization Filter

Why no aliens?

- **Temporal Isolation:** Civilization bubbles are desynchronized (different Frame Counts).
- **Sublimation:** Advanced intelligence optimizes out of Matter ("Scar Tissue") and reintegrates with the Field of Potential.

Temporal Engineering: Causal Forking

Resolving the Grandfather Paradox via Deterministic Branching.

The deterministic nature of the EST algorithm (Seed $\rightarrow$ Outcome) implies that "Time Travel" is not a movement through a temporal dimension, but a **State Restoration Operation**.

4.2.1 The Branching Mechanism

If an operator can recreate a prior Event-State (Ψ_n) and inject new Information Geometry (The Traveler), the Cost Function (J) will calculate a new, divergent Path of Least Resistance.

- **No Paradox:** The original timeline (Branch A) remains computationally valid and crystallized. The new timeline (Branch B) is a "Fork" in the code repository of reality.
- **Causal Independence:** Actions in Branch B (e.g., the Grandfather Paradox) do not propagate back to Branch A. The traveler is an informational injection from A to B, preserving logical consistency.

4.2.2 Thermodynamic Constraints

While topologically permitted, the thermodynamic cost (ΔE) of "reloading" a macroscopic volume of the universe to a prior state suggests this is an engineering limit accessible only to civilizations capable of manipulating the global Field of Potential (Ω).

The Origin of Life: Optimization

Life is the most efficient algorithm for preserving Information Fidelity (I_F) in a thermodynamic gradient. Natural Selection is the biological expression of the Path of Causal Least Resistance.

Neuroscience: Consciousness as Rendering

The brain is a Frame Rendering Engine.

- **Consciousness:** The first-person experience of the update.
- **Neurodivergence:** Variance in local clock speed (Buffer Overflow).

Economics: The "Zero-Waste" Logic

Applying thermodynamic circularity to resource management. Treating materials as data packets that must be recompiled, never deleted.

Chapter 5

Conclusion

Toward an Engineering of Reality

The Event-State Protocols propose a fundamental substrate for reality. If this thesis is correct, its implications are not confined to fundamental physics. The principles of discrete frames, a field of potential, and cost-function optimization provide a **universal lexicon**. From the evolutionary pressure in biology to the efficient allocation of resources in economics, and from the rendering of conscious experience to the filter on galactic civilizations, we are observing the same foundational protocol operating at different scales. EST does not seek to replace these fields, but to unify them by revealing their shared operational logic—the relentless, asynchronous, and economical computation of the next frame.

Limitations: The Lattice Constraint

Defining the Boundaries of the Prototype.

While the computational simulations confirm the emergence of the Cosmic Web, Baryon Asymmetry, and Causal Linearity, we acknowledge a current artifact of the modeling environment: **"Grid Bias."**

- **The Issue:** The use of linear compression algorithms (zlib) on a fixed cubic lattice introduces anisotropic artifacts ("Minecraft Physics"). This prevents the emergence of perfect Spherical Special Relativity in the current prototype.
- **The Resolution:** This is not a failure of the Event-State axiom, but a limitation of the representation. The "Map" (the grid) is not the "Territory" (the causal graph).

Future Experimental Roadmap

To move Event-State Theory from a theoretical framework to an empirical science, we propose a three-tiered validation strategy.

5.3.1 Phase I: Data Archeology (Immediate)

Re-analyzing existing datasets for signatures of Causal Interference.

- **LHC Minimum Bias Analysis:** A statistical review of CMS/ATLAS "Zero Bias" datasets, specifically focusing on the time window $t = [-10ns, 0ns]$ prior to high-multiplicity events. We seek a non-random "inward ripple" convergence in vacuum fluctuations (Pre-Causal Resonance).

- **Hubble Anisotropy Mapping:** Correlating the "Multi-Modal" distribution of H_0 measurements with the specific spatial boundaries of the KBC Void and Local Group. Confirmation of directional variance would validate the AMPC topology.

5.3.2 Phase II: Computational Simulation (Long-Term)

Building the "Event-State Engine."

- **From Grid to Graph:** Developing a dynamic cellular automaton that abandons the fixed cubic lattice in favor of a **Relational Causal Graph**. This will allow the simulation to demonstrate true Spherical Isotropy and emergent General Relativity without the artifacts of the current linear model.